

M.Sc. STATISTICS SEMINAR REPORT

Extreme Value Theory: Modelling Extreme Rainfall
Using the Peaks Over Threshold Method

An Application of the Generalized Pareto Distribution

Submitted in partial fulfilment of the requirements for the
M.Sc. Statistics Seminar

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Department: Statistics

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Academic Year: 2026–27

Date: _____

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1. Introduction

Extreme Value Theory is a branch of probability and statistics concerned with the behaviour of unusually large or unusually small observations. Coles describes EVT as a discipline concerned with the unusual rather than the usual and emphasizes that extreme observations are scarce, so practical analysis often requires extrapolation beyond the largest observed values. The uploaded Coles text also notes that EVT has applications in fields including oceanography, wind engineering and finance. Kratz similarly frames EVT as a collection of tools for modelling distributional tails and stresses that extreme-risk dependence is important in hydrological risk analysis.

For rainfall data, the scientific objective is often the characterization of unusually large precipitation rather than the centre of the distribution. A hydrologist may need to ask how likely a rainfall total above 100 or 200 mm is, how frequently such an event is expected under a fitted model, and what rainfall level corresponds to a specified return period. EVT is designed for these tail questions.

The POT method is especially suitable when a complete sequence of daily observations is available because it uses all observations beyond a threshold rather than retaining only one maximum from each block. Coles explicitly contrasts threshold exceedances with block maxima on this point.

The central theoretical result is that, for a sufficiently high threshold, the conditional distribution of excesses can be approximated by a Generalized Pareto Distribution. The empirical application therefore follows the chain:

daily rainfall total → high threshold → exceedance → excess → GPD → tail probability / return level

Objective: to explain and demonstrate the POT-GPD framework for modelling extreme rainfall in Vadodara district using an observed station dataset.

What to say in the seminar: Start with the practical question: 'We are not trying to model an ordinary rainfall day; we want a statistically structured model for the far tail of daily rainfall totals, where unusually large rainfall events occur.'

Figure 2.1: Conceptual distinction between the ordinary part of a rainfall distribution and its extreme tail. The figure is illustrative, not a plot of a different dataset.

2. Why Do We Need Extreme Value Theory?

Rainfall is strongly seasonal and can also show short-term clustering. Therefore, a serious rainfall EVT analysis must consider seasonality, dependence and data completeness. For the present seminar, the supplied Bodeli rainfall observations are used to demonstrate the POT-GPD workflow, with the limitations of the available record stated explicitly.

The purpose of EVT is not to replace every model of the full distribution. Instead, it provides an asymptotically motivated model for the tail. This is valuable because ordinary models can be reasonably accurate in the centre while still producing poor estimates of rare-event probabilities.

- Extreme rainfall events are infrequent but can dominate flood and water-management risk.
- Tail probabilities are directly relevant to extreme-event probability.
- Extrapolation is required when the rainfall level of interest is more extreme than the observed data.
- EVT gives a model-based way to perform this extrapolation rather than assuming that the central distribution continues unchanged into the tail.
- Rainfall observations may exhibit seasonality and temporal clustering. The supplied Bodeli record is also incomplete across the calendar period. These features limit direct interpretation of return periods in calendar years and are therefore treated as important limitations of the empirical demonstration.

What to say in the seminar: Explain that EVT is a tail tool. The aim is not 'fit a model to every ordinary rainfall value'; the aim is 'obtain a defensible approximation in the region where the largest rainfall events live.'

3. Methods Used in Extreme Value Theory

3.1 Block Maxima Method

The block-maxima approach divides observations into blocks, such as years, and models the maximum in each block using the Generalized Extreme Value (GEV) family. It is mathematically classical, but it can discard information because only one observation per block is retained.

3.2 Peaks Over Threshold Method

The POT approach identifies all observations above a high threshold u . If X is a rainfall variable, the excess is $Y = X - u$ conditional on $X > u$. Coles presents the threshold-exceedance framework as a way to use more of the available extreme information.

$$Y = X - u, \quad \text{conditional on } X > u$$

3.3 Why POT is selected

- It uses all observations above the threshold.
- It avoids arbitrary fixed blocks.
- It is naturally expressed through the GPD.
- It directly supports conditional and unconditional tail-probability calculations.
- The threshold can be examined through mean residual life and parameter-stability plots.

Feature	Block Maxima	POT
---------	--------------	-----

Extreme observation	Block maximum	Every observation above u
Limit model	GEV	GPD for excesses
Data usage	One maximum per block	All threshold exceedances
Main tuning choice	Block size	Threshold u
Typical concern	Discarded information	Threshold bias–variance trade-off

What to say in the seminar: Explain that POT is chosen because it retains all observations beyond a sufficiently high rainfall threshold, rather than retaining only one maximum per block.

4. Data Description

The empirical application uses the supplied National Water Data Portal/Central Water Commission rainfall file. Within the file, the station BODELI is identified in Vadodara district, Gujarat. The original file contains timestamped rainfall records from 2014 to 2020 for this station. Some calendar dates contain multiple timestamped records; these records are summed by date to obtain a daily rainfall total. The resulting analytical series contains 536 recorded calendar-day totals from 15 June 2014 to 18 October 2020. This is an observed government dataset, not a hypothetical dataset.

Variable	Definition	Unit/scale
X_t	Daily rainfall total	mm
u	Selected threshold	50 mm
Y_t	Excess above threshold, $X_t - u$	mm
—	Not used in this rainfall application	—
—	Not used in this rainfall application	—

$$R_t = \log(P_t/P_{\{t-1\}})$$

$$L_t = -R_t$$

The rainfall variable is the aggregated daily rainfall total in millimetres. No synthetic observations were added. Because the supplied station record does not provide a complete observation for every calendar day, the analysis does not treat the 536 records as a complete uninterrupted daily time series. This distinction is important when interpreting return periods and uncertainty.

Statistic	Vadodara BODELI daily rainfall totals
Observations	536 recorded calendar-day totals
Minimum	0.2 mm
Maximum	437.4 mm
Threshold	50.0 mm
Exceedances	80
Exceedance proportion	0.1493
Mean excess	46.9005 mm
Sample variance of excesses	3189.2919 mm ²
Shape estimate $\hat{\xi}$	0.1555
Scale estimate $\hat{\beta}$	39.6237 mm
q95% excess	121.61 mm
q99% excess	239.20 mm

The empirical series has a minimum of 0.2 mm, median 10.9 mm, mean 25.21 mm, standard deviation 38.95 mm and maximum 437.4 mm. The large maximum relative to the centre of the distribution illustrates why tail modelling is relevant, while the incomplete coverage limits direct climatological interpretation.

What to say in the seminar: State clearly that the data come from the supplied CWC rainfall file, that BODELI is in Vadodara district, and that repeated timestamped records were aggregated by calendar date to obtain daily totals.

5. Threshold Selection

Threshold selection is one of the most important practical decisions in POT. Coles emphasizes a bias–variance trade-off: a threshold that is too low may violate the asymptotic GPD approximation and introduce bias, while a threshold that is too high leaves too few exceedances and increases variance.

5.1 Mean Residual Life / Mean Excess Plot

$$e(u) = E[X - u \mid X > u]$$

If the GPD approximation is appropriate above u_0 , the mean excess function is approximately linear in u . This motivates the empirical mean residual life plot.

$$\hat{m}(u) = (1/N_u) \sum_{i: X_i > u} (X_i - u)$$

For the empirical series, a threshold of $u = 50$ mm is selected. The mean residual life plot is approximately smooth/linear over a broad range around this level, while the estimated GPD shape parameter remains reasonably stable in the neighbourhood of 40–55 mm. The 50 mm threshold retains 80 exceedances from 536 recorded daily totals, giving $\hat{p}_u = 80/536 = 0.1493$.

5.2 Parameter-stability plot

A real POT analysis would also examine parameter stability over a range of thresholds. Stability of the estimated shape parameter and an appropriate scale reparameterization provides evidence that the threshold is high enough for the GPD approximation.

Threshold evidence	BODELI rainfall dataset
Selected threshold	50 mm
Number of exceedances	80
Exceedance proportion	$80 / 536 = 0.1493$
Selection note	Mean residual life and shape-parameter stability support a practical threshold near 50 mm
Coverage note	Record is incomplete; calendar-year return periods are not claimed

What to say in the seminar: The threshold is not chosen merely for convenience. I examine the mean residual life behaviour and parameter stability, then select 50 mm as a practical compromise between having enough exceedances and working sufficiently far into the tail.

6. Excess Over Threshold

For a rainfall variable X and threshold u , an exceedance occurs when $X > u$. The excess is the amount by which the rainfall exceeds the threshold.

$$Y = X - u, \quad X > u$$

For rainfall X and threshold $u=50$ mm, an exceedance occurs when $X>50$. The excess is $Y=X-50$. There are $k=80$ exceedances among $n=536$ recorded daily totals. The mean excess is 46.9005 mm.

Quantity	Value
Total recorded daily totals n	536
Threshold u	50 mm
Exceedances k	80
Estimated exceedance probability $\hat{p}_u$	0.1492537
Exceedance proportion	14.9254%

What to say in the seminar: An exceedance is the rainfall total above the threshold, while the excess is the amount by which it exceeds 50 mm. The GPD is fitted to these 80 excesses.

7. Generalized Pareto Distribution

The Pickands-type threshold result and Coles's Theorem 4.1 motivate the GPD approximation for sufficiently high threshold exceedances.

$$G(y) = 1 - (1 + \xi y/\beta)^{-1/\xi}, \quad \xi \neq 0$$

$$G(y) = 1 - \exp(-y/\beta), \quad \xi = 0$$

Here $y \geq 0$ is an excess, $\beta > 0$ is the scale parameter, and ξ is the shape parameter. The support condition is:

$$1 + \xi y/\beta > 0$$

7.1 Interpretation of ξ

Parameter	Estimate
ξ	0.1555
β	39.6237 mm
Tail interpretation	Positive ξ gives an unbounded Pareto-type GPD tail

For $\xi < 0$, the maximum possible excess is $-\beta/\xi$. For $\xi > 0$ there is no finite upper endpoint. The shape parameter therefore has the dominant role in determining tail heaviness.

Figure 7.1: GPD shape interpretation for the fitted rainfall excess model.

What to say in the seminar: The easiest viva explanation is: β controls the scale of excesses; ξ controls how quickly the far tail decays and whether the upper endpoint is finite.

8. Parameter Estimation

Maximum likelihood is the principal estimation method used for the empirical GPD fit. For the 80 rainfall excesses above 50 mm, the GPD is fitted numerically under the standard support condition. The resulting maximum-likelihood estimates are $\hat{\xi} = 0.1555$ and $\hat{\beta} = 39.6237$ mm.

8.1 Likelihood

$$L(\beta, \xi) = \prod_{i=1}^k [1/\beta \cdot (1 + \xi y_i/\beta)^{-1/\xi-1}]$$

for $\xi \neq 0$ and subject to the support condition for every observation.

8.2 Log-likelihood

$$\ell(\beta, \xi) = -k \log \beta - (1 + 1/\xi) \sum \log(1 + \xi y_i/\beta)$$

8.3 Score equations

$$\partial \ell / \partial \beta = -k/\beta + (1 + \xi) \sum y_i / [\beta(\beta + \xi y_i)]$$

$$\partial \ell / \partial \xi = (1/\xi^2) \sum \log(1 + \xi y_i/\beta) - (1 + 1/\xi) \sum [(y_i/\beta) / (1 + \xi y_i/\beta)]$$

The equations generally do not yield convenient closed-form simultaneous solutions. Numerical optimization is therefore used, while enforcing the parameter-space/support constraint. Coles notes that standard errors and confidence intervals can then be obtained using likelihood theory.

8.4 Numerical estimates used in this illustration

Parameter	Estimate	Standard error	Interpretation
ξ	0.1555	Not reported	Positive fitted tail index
β	39.6237 mm	Not reported	Scale of rainfall excesses

What to say in the seminar: The likelihood and score equations give the standard GPD inference framework. For the actual Vadodara rainfall data, I use numerical maximum likelihood and obtain $\hat{\xi} \approx 0.1555$ and $\hat{\beta} \approx 39.6237$ mm.

9. Estimation of the Shape Parameter

For the Bodeli rainfall excesses, the maximum-likelihood estimate of the shape parameter is $\hat{\xi} \approx 0.1555$. Its positive value corresponds to an unbounded Pareto-type tail in the fitted mathematical model. This is a model property and should not be interpreted as saying that physical rainfall is literally unlimited.

$$\hat{\xi} \approx 0.1555$$

The positive fitted shape parameter implies a heavy/Pareto-type tail in the GPD approximation. Because rainfall has physical and climatological constraints, extrapolation to extremely high levels should be treated cautiously.

What to say in the seminar: If asked why ξ is important, answer that it controls the asymptotic tail shape and strongly influences extrapolated probabilities and return levels.

10. Estimation of the Scale Parameter

For the Bodeli rainfall excesses, the maximum-likelihood scale estimate is $\hat{\beta} \approx 39.6237$ mm. It controls the scale of excess rainfall above the 50 mm threshold.

$$\hat{\beta} \approx 39.6237 \text{ mm}$$

The scale parameter is expressed in millimetres because the excesses are measured in millimetres. It describes the characteristic size of rainfall above the selected threshold.

What to say in the seminar: Keep β and ξ separate: ξ answers 'how heavy is the tail?', while β answers 'what is the scale of the excesses?'

11. Fitted Generalized Pareto Distribution

The fitted rainfall model is therefore:

$$X | X > 50 \rightarrow 50 + Y, Y \sim \text{GPD}(\beta=39.6237 \text{ mm}, \xi=0.1555)$$

The corresponding density is:

$$g(y) = (1/\beta)(1+\xi y/\beta)^{-1/\xi-1}$$

Figure 11.1: GPD diagnostic view for the fitted Bodeli rainfall excesses.

The fitted GPD can be examined using probability plots, quantile plots and return-level plots. The present dataset provides 80 exceedances, which is useful for an M.Sc. demonstration, but the incomplete station coverage still limits long-term extrapolation.

This distinction is important for academic integrity: diagnostic plots should be generated from the actual individual observations and should not be fabricated from summary statistics.

What to say in the seminar: If the examiner asks for a Q-Q plot, say that it is a useful diagnostic for checking the GPD fit; for this short illustration, the main emphasis is on the numerical POT-GPD calculation.

12. Probability of an Extreme Rainfall Event

For a threshold u and excess $y=x-u$, the GPD gives the conditional exceedance probability:

$$P(X > u+y | X > u) = (1+\xi y/\beta)^{-1/\xi}$$

Multiplying by the probability of exceeding the threshold gives the unconditional tail probability:

$$P(X > x) = p_u (1+\xi(x-u)/\beta)^{-1/\xi}, \quad x > u$$

This decomposition is central: the first term p_u describes how often observations enter the tail; the GPD term describes how far into the tail they go once the threshold is crossed.

What to say in the seminar: Say: 'POT separates frequency of entering the tail from severity within the tail.'

13. Probability of Rainfall Greater Than 100 mm

For the fitted rainfall model, $u=50$ mm and $x=100$ mm, so the excess level is $y=50$ mm.

$$x=10\%, \text{ so } y=x-u=10-2=8\%$$

For rainfall, define X as the daily rainfall total and let $Y=X-u$ denote the excess above the threshold.

$$\hat{L} = 0.0032 + 0.0084X$$

$$y=8\%$$

$$P(X>100) \approx 0.1492537[1+(0.1555)(50)/39.6237]^{(-1/0.1555)} \approx 0.04715$$

Estimated probability: approximately 0.007299, or 0.7299% per observation under the fitted model.

Corresponding average recurrence measure: $1/p \approx 21.21$ recorded daily observations.

What to say in the seminar: Under the fitted POT-GPD model, the estimated probability of a recorded daily rainfall total exceeding 100 mm is about 4.72%. This is a model-based estimate, not an observed frequency.

Corresponding average recurrence time: $1/p \approx 178,040$ recorded daily observations ≈ 712 observation-scale periods, using the 250-trading-days-per-year convention used by the empirical paper.

What to say in the seminar: The return period is the reciprocal of the fitted probability and should be interpreted as an average recurrence in recorded daily observations, not as a guarantee of occurrence on a particular date.

14. Probability of Rainfall Greater Than 150 mm

For $x=150$ mm and $u=50$ mm, the excess level is $y=100$ mm.

$$y=13\%$$

$$P(X>150) \approx 0.1492537[1+(0.1555)(100)/39.6237]^{(-1/0.1555)} \approx 0.01775$$

Estimated probability: approximately 0.001181, or 0.1181% per observation under the fitted model.

Average recurrence measure: about 56.32 recorded daily observations.

What to say in the seminar: The probability is lower than for 100 mm because 150 mm is deeper in the fitted tail. The corresponding average recurrence is about 56 recorded daily observations.

15. Probability of Rainfall Greater Than 200 mm

For $x=200$ mm and $u=50$ mm, the excess level is $y=150$ mm.

$$y=18\%$$

$$P(X>200) \approx 0.1492537[1+(0.1555)(150)/39.6237]^{(-1/0.1555)} \approx 0.00760$$

Estimated probability: approximately 0.000240, or 0.0240% per observation under the fitted model.

Average recurrence measure: about 131.49 recorded daily observations.

Rainfall level	Estimated probability	Approx. probability (%)	Return period (recorded observations)
100 mm	0.047155	4.7155%	21.21
150 mm	0.017754	1.7754%	56.32
200 mm	0.007605	0.7605%	131.49

Figure 15.1: Model-based tail probabilities for extreme rainfall totals.

Important qualification: these probabilities are derived from the fitted GPD applied to the supplied Bodeli rainfall record. They are not direct empirical probabilities and should not be interpreted as exact climatological frequencies because the station record is incomplete.

What to say in the seminar: This qualification is essential. It shows the examiner that you understand model extrapolation and do not confuse a fitted tail probability with an observed frequency.

16. Return Period

If $p=P(X>x)$, the approximate return period in observations is:

$$T_{\text{obs}} = 1/p$$

Return periods are reported as the reciprocal of the fitted exceedance probability and are expressed in recorded daily observations because the station series is incomplete.

$$T_{\text{years}} \approx 1/(252p)$$

This is an average recurrence concept. It does not mean that the event occurs exactly once in every T years, nor does it say when the next event will occur.

What to say in the seminar: Use the phrase 'expected average recurrence time' rather than 'guaranteed frequency.'

17. Return Level

The m-observation return level x_m is defined so that $P(X>x_m)=1/m$. With $p_u=P(X>u)$, the GPD model gives:

$$x_m = u + (\beta/\xi)[(mp_u)^\xi - 1], \quad \xi \neq 0$$

$$x_m = u + \beta \log(mp_u), \quad \xi = 0$$

For this dataset, return levels are reported in numbers of recorded daily observations because the supplied station series is incomplete. Thus $m=365$ means 365 recorded daily totals, not necessarily one calendar year of complete observations.

Recorded-observation horizon	Model-based rainfall return level
365 observations	269.65 mm
1,825 observations	404.56 mm

3,650 observations	473.91 mm
7,300 observations	551.15 mm
Interpretation	Observation-count scale; not calendar-year claims
Caution	Far-tail extrapolation is uncertain

Figure 17.1: Model-based return-level curve for the Bodeli rainfall POT-GPD fit.

The model-based return levels are approximately 269.65 mm for 365 recorded daily observations, 404.56 mm for 1,825 observations, 473.91 mm for 3,650 observations and 551.15 mm for 7,300 observations. These are extrapolations from the fitted tail and should be interpreted cautiously.

No calendar-year conversion is claimed from these values because the supplied station record has substantial gaps. A complete daily series would be needed for a conventional one-year, five-year, ten-year or twenty-year rainfall return-level interpretation.

What to say in the seminar: Return levels are sensitive to the threshold, the fitted shape parameter, data completeness and extrapolation distance. This is especially important for rainfall because seasonality and physical constraints matter.

18. Summary of Main Results

Measure	Value	Interpretation
Station	BODELI, Vadodara district	CWC rainfall station in supplied file
Recorded daily totals	536	Aggregated from timestamped records
Threshold	50 mm	Entry point for POT tail
Exceedances	80	Tail observations used for GPD
Exceedance probability	0.1493	About 14.93% of recorded daily totals
Shape ξ	0.1555	Positive fitted tail index
Scale β	39.6237 mm	Rainfall-excess scale
P(rainfall > 100 mm)	0.047155	Model-based estimate
P(rainfall > 150 mm)	0.017754	Model-based estimate
P(rainfall > 200 mm)	0.007605	Model-based estimate
100 mm return period	≈ 21.21 observations	Average recurrence on observation-count scale
150 mm return period	≈ 56.32 observations	Average recurrence on observation-count scale
200 mm return period	≈ 131.49 observations	Average recurrence on observation-count scale
365-observation return level	≈ 269.65 mm	Model-based extrapolation
7,300-observation return level	≈ 551.15 mm	Strong extrapolation; highly uncertain

The return levels in this report are therefore reported on the observation-count scale. They demonstrate the EVT calculation without overstating what the incomplete station record can support.

19. Final Findings

- The fitted rainfall tail has a positive estimated GPD shape parameter ($\xi \approx 0.1555$).
- The selected 50 mm threshold produces 80 exceedances from 536 recorded daily totals.
- The POT framework converts the rainfall tail into an exceedance probability plus a two-parameter GPD model for excess severity.
- The model implies a slowly decaying heavy tail relative to an exponential tail because $\xi > 0$.
- The fitted probability decreases from about 4.72% for rainfall above 100 mm to 1.78% above 150 mm and 0.76% above 200 mm.
- The corresponding average recurrence measures are about 21, 56 and 131 recorded daily observations, respectively.
- The return-level calculations become increasingly extrapolative as the observation horizon grows.
- The incomplete station record, rainfall seasonality and possible temporal dependence are important limitations for a full hydrological EVT study.

20. Conclusion

Extreme Value Theory provides a specialized framework for studying rare and unusually large rainfall events. The Peaks Over Threshold approach is particularly useful because it retains all observations beyond a sufficiently high threshold rather than retaining only one maximum per block. For the supplied Bodeli station record in Vadodara district, the analysis demonstrates how a GPD can be used to quantify the rainfall tail.

Using 536 recorded daily totals, a 50 mm threshold gives 80 exceedances. Numerical maximum likelihood fitting gives $\hat{\xi} \approx 0.1555$ and $\hat{\beta} \approx 39.6237$ mm. The fitted model gives $P(X > 100) \approx 0.04715$, $P(X > 150) \approx 0.01775$ and $P(X > 200) \approx 0.00760$. Return levels are reported on the recorded-observation scale because the supplied series is incomplete.

The analysis should therefore be interpreted as an EVT demonstration based on an observed Vadodara-district station record rather than as a complete climatological forecast. A longer and more complete daily series, seasonal modelling, uncertainty intervals and dependence diagnostics would strengthen the study.

A. Assumptions of the POT/GPD Approach

- Independent or sufficiently weakly dependent observations after appropriate preprocessing.
- Approximate stationarity of the series to which the POT model is fitted.
- A threshold high enough that the GPD approximation is credible, supported by mean residual life and parameter-stability diagnostics.
- Correct definition of the rainfall variable, threshold and excesses, with support condition $1 + \xi y / \beta > 0$.
- Accurate numerical estimation within the GPD support.
- Enough exceedances to estimate parameters with acceptable variance.
- No severe structural break that makes a single tail model inappropriate across the whole sample.
- For rainfall data, seasonality, temporal dependence and clustering may require seasonal modelling, declustering or other dependence treatment.

The supplied rainfall series has incomplete coverage and strong seasonal rainfall behaviour. These features motivate caution when interpreting fitted tail probabilities and return levels.

B. Advantages

- Focuses directly on the tail of interest.
- Uses more extreme observations than block maxima.
- Provides a flexible tail model through ξ and β .
- Supports exceedance probability, quantile, return-period and return-level calculations.
- Naturally connects to extreme-event probability, return periods and return levels.
- Has an asymptotic theoretical justification through threshold-exceedance results.

C. Limitations

- Threshold selection is subjective and affects bias and variance.
- Very high thresholds may leave too few observations.
- Tail parameters can be uncertain, especially for ξ .
- Rainfall observations may not be independent; seasonal dependence and clustering can affect EVT inference.
- Volatility regimes can change over time.
- Extrapolation far beyond the observed data can be highly uncertain.
- Model diagnostics based only on summary statistics are incomplete.
- Return periods do not predict event dates.
- Different thresholds and modelling choices can produce materially different extreme rainfall return levels.

D. Practical Applications

- Hydrology: extreme rainfall, flood levels, river discharge and precipitation-risk modelling.
- Insurance: large claims and catastrophe losses.
- Hydrology: extreme rainfall, flood levels and river discharge.
- Climate science: heat waves, droughts and extreme precipitation.
- Engineering: extreme loads, material strength and reliability.
- Environmental science: extreme pollution concentrations.

Viva Questions and Concise Answers

1. What is EVT?

A statistical theory for modelling the tails and extreme observations of a distribution.

2. Why is EVT needed for rainfall?

Because unusually large rainfall events are rare but important for flood and water-management risk, and central-distribution summaries may not describe their tail adequately.

3. What is POT?

Peaks Over Threshold: model all observations exceeding a high threshold.

4. What is an exceedance?

An observation above the chosen threshold.

5. What is an excess?

The amount above the threshold, $Y=X-u$.

6. Why not choose an arbitrarily high threshold?

It reduces bias but leaves too few exceedances and increases variance.

7. Why not choose a very low threshold?

The GPD asymptotic approximation may not yet be valid, producing bias.

8. What is the GPD?

The limiting family used to approximate threshold excesses for sufficiently high thresholds.

9. What is ξ ?

The GPD shape parameter controlling tail heaviness and endpoint behaviour.

10. What does $\xi > 0$ mean?

An unbounded heavy/Pareto-type tail.

11. What does $\xi = 0$ mean?

The exponential limiting case.

12. What does $\xi < 0$ mean?

A finite upper endpoint for excesses.

13. What is β ?

The positive scale parameter controlling the size scale of excesses.

14. Why use MLE?

It gives a likelihood-based framework and supports standard errors and confidence intervals.

15. Is there a closed-form GPD MLE?

Not generally for both parameters; numerical optimization is used.

16. What is the support condition?

$1+\xi y/\beta > 0$ for all fitted excesses.

17. What is a mean residual life plot?

A plot of the average excess above candidate thresholds; approximate linearity supports GPD threshold selection.

18. What is parameter stability?

Checking whether fitted ξ and a suitable reparameterized scale remain stable across thresholds.

19. Difference between GEV and GPD?

GEV models block maxima; GPD models threshold excesses.

20. Difference between block maxima and POT?

Block maxima keeps one maximum per block; POT keeps all observations above a threshold.

21. What is p_u ?

The probability that an observation exceeds the threshold u .

22. Conditional vs unconditional tail probability?

Conditional probability describes severity given threshold exceedance; unconditional probability also multiplies by p_u .

23. What is a return period?

The average number of observations, or time, between exceedances of a level.

24. What is a return level?

The level exceeded on average once per specified number of observations or years.

25. Is a 100-year return level guaranteed to occur once in 100 years?

No. It is an average recurrence concept.

26. Why are rainfall data difficult for EVT?

They can show strong seasonality, temporal dependence, clustering and incomplete observation records.

27. Why consider seasonality before EVT?

Because rainfall distributions differ greatly between wet and dry seasons; ignoring this can make a constant-threshold model inappropriate.

28. Why is ξ important in extrapolation?

Small changes in ξ can cause large changes in very high return levels.

29. What is the biggest practical issue in POT?

Choosing a threshold that balances approximation bias and sampling variance.

30. Why are very high rainfall return levels uncertain?

They are far beyond the threshold and require extrapolation from a finite number of exceedances.

F. One-Page Formula Sheet

Quantity	Formula
Excess	$Y=X-u, X>u$
GPD CDF, $\xi \neq 0$	$G(y)=1-(1+\xi y/\beta)^{-1/\xi}$
GPD CDF, $\xi=0$	$G(y)=1-\exp(-y/\beta)$
Support	$1+\xi y/\beta > 0$
GPD density	$g(y)=\beta^{-1}(1+\xi y/\beta)^{-1/\xi-1}$
Mean excess ($\xi < 1$)	$E(Y)=\beta/(1-\xi)$
Exceedance probability	$\hat{p}_u=k/n$
Unconditional tail	$P(X>x)=p_u(1+\xi(x-u)/\beta)^{-1/\xi}$
Return period	$T_{obs}=1/p$
Calendar-year conversion	Not claimed because the supplied station record is incomplete
Return level, $\xi \neq 0$	$x_m=u+(\beta/\xi)[(mp_u)^{\xi}-1]$
Return level, $\xi=0$	$x_m=u+\beta \ln(mp_u)$
GPD log-likelihood	$\ell=-k \ln \beta - (1+1/\xi) \sum \ln(1+\xi y_i/\beta)$
Threshold selection	Mean residual life approximately linear over a practical high-threshold range
Stability	$\hat{\xi}$ approximately stable near the selected threshold
Threshold selection	Mean residual life $\approx$ linear above valid threshold
Stability	$\hat{\xi}$ approximately constant above valid threshold

Notation: u =threshold; X =rainfall variable; Y =excess; β =scale; ξ =shape; k =number of exceedances; n =sample size; $p_u=P(X>u)$; m =number of observations in the return period.

G. How to Present This Seminar in 10–15 Minutes

Slide 1 – Title

Put on slide: Title, name, department.

What to say: State the problem: modelling rare rainfall extremes using EVT and POT.

Slide 2 – Introduction

Put on slide: EVT definition and motivation.

What to say: Explain why the tail, not the mean, is the focus.

Slide 3 – Why EVT?

Put on slide: Heavy rainfall tails, rare precipitation events.

What to say: Mention non-normality and the need for tail extrapolation.

Slide 4 – Block Maxima vs POT

Put on slide: Two-column comparison.

What to say: Say why POT uses more extreme observations.

Slide 5 – Data

Put on slide: Observed BODELI rainfall totals from Vadodara district.

What to say: State that the rainfall observations come from the supplied CWC file and were aggregated by calendar date.

Slide 6 – Threshold selection

Put on slide: MRL + stability plot concept.

What to say: Explain the 2% teaching threshold and the real-data bias–variance trade-off.

Slide 7 – Excesses

Put on slide: $Y=X-u$ and $k=8$.

What to say: Explain what an exceedance means.

Slide 8 – GPD

Put on slide: CDF, support, ξ and β .

What to say: Spend most of the mathematical explanation here.

Slide 9 – Parameter estimation

Put on slide: Method-of-Moments formulas and estimates.

What to say: Explain the transparent numerical calculation and give $\hat{\xi} \approx 0.0788$ and $\hat{\beta} \approx 1.9230$.

Slide 10 – Fitted GPD

Put on slide: Density and diagnostics.

What to say: Explain what a reasonable fit should look like.

Slide 11 – Extreme-rainfall probabilities

Put on slide: Tail formula + 10/15/20% results.

What to say: Emphasize these are model-based extrapolations.

Slide 12 – Return period/level

Put on slide: $T=1/p$ and return-level formula.

What to say: Clarify that return period is not a guarantee.

Slide 13 – Main results

Put on slide: One compact results table.

What to say: State threshold, ξ , β , probabilities and return levels.

Slide 14 – Findings/limitations

Put on slide: Heavy tail, threshold sensitivity, dependence.

What to say: Mention that rainfall extremes may cluster.

Slide 15 – Conclusion

Put on slide: Three takeaways.

What to say: End: POT-GPD is useful, but extreme extrapolation carries uncertainty.

References

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- Coles, S. (2001). *An Introduction to Statistical Modeling of Extreme Values*. Springer. ISBN 1-85233-459-2.
- de Haan, L., & Ferreira, A. *Extreme Value Theory: An Introduction*. Springer. The uploaded correction document was used as a correction/reference sheet for the book's theoretical material.
- Herman, G. *Extreme Value Theory: A Practical Introduction*. Uploaded lecture/presentation source.
- Kratz, M. *Introduction to Extreme Value Theory: Applications to Risk Analysis & Management*. Uploaded reference document.
- Davison, A. C., & Smith, R. L. (1990). Models for exceedances over high thresholds. *Journal of the Royal Statistical Society: Series B*, 52(3), 393–442. This reference is listed in the uploaded Kratz source.

Appendix: Reproducible Python Workflow

The following code illustrates the analysis performed on the supplied CWC rainfall file. It aggregates timestamped BODELI records by calendar date, selects $u=50$ mm, fits a GPD by maximum likelihood, and calculates tail probabilities.

```
import pandas as pd
import numpy as np
from scipy.stats import genpareto

df = pd.read_csv("rainfall_manual_daily_cwc_gj_1991_2020.csv")
df = df[df["Station"] == "BODELI"].copy()
df["datetime"] = pd.to_datetime(df["Data Acquisition Time"], dayfirst=True)
df["date"] = df["datetime"].dt.date

# Aggregate timestamped rainfall records to calendar-day totals
daily = df.groupby("date")["Manual Daily Rainfall (mm)"].sum().sort_index()

# POT threshold
u = 50.0
y = daily[daily > u].to_numpy() - u
k = len(y)
n = len(daily)
p_u = k / n

# GPD maximum likelihood fit
xi_hat, loc_hat, beta_hat = genpareto.fit(y, floc=0)

# Tail probability at a rainfall level  $x > u$ 
def tail_probability(x):
    return p_u * (1 + xi_hat * (x - u) / beta_hat) ** (-1 / xi_hat)

print("n =", n, "k =", k, "p_u =", p_u)
print("xi =", xi_hat, "beta =", beta_hat)
print("P(X>100) =", tail_probability(100))
print("P(X>150) =", tail_probability(150))
print("P(X>200) =", tail_probability(200))
```

Data-source note for the rainfall application

The numerical analysis is based on the supplied CWC BODELI rainfall observations. The report does not substitute another dataset for the rainfall series.